

TriangleFX

From Three Equations to a Valid Triangle

Objective

Create a JavaFX application that takes three equations, finds where the lines meet, checks if they form a valid triangle, and displays the triangle on the screen.

Group 1

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Course

CSC360 Computer Graphics and Digital
Image Processing

1. Problem Statement

The Problem

We are given **three equations** representing three lines. Our goal is to find where the lines meet and determine whether they form a valid triangle.

What the system does:

- Takes three equations as input
- Finds the three intersection points
- Checks if the intersection points form a triangle
- Calculates area, perimeter, sides and centroid
- Draws the triangle on a coordinate plane

When is it Invalid?

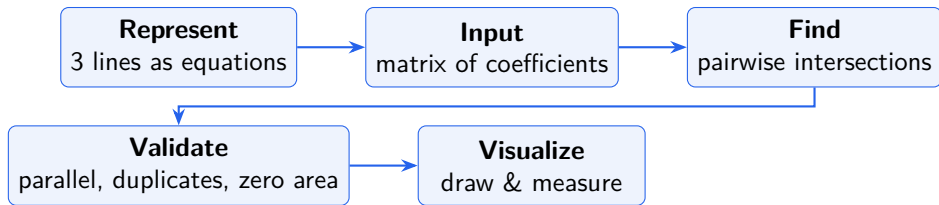
- Two lines are parallel
- Two lines are the same
- All three lines meet at one point
- The three intersection points lie on one line
- Input is missing or invalid

Example

$$x + y = 8, x - y = 2, x = 1$$

$$\Rightarrow \text{vertices } (5, 3), (1, -1), (1, 7)$$

2. Thought Process



Principles

- Represent the three lines using a structured **3×2 matrix + 3 constants**
- Keep mathematical logic **separate from the UI**
- Treat invalid geometry as an expected outcome

How We Built It

- Support direct matrix input and predefined presets
- Solve each pair of lines independently
- Automatically scale the canvas to display the triangle

3. The Mathematics

1. Matrix Formulation ($Ax = b$)

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad \begin{array}{l} \text{Line 1} \\ \text{Line 2} \\ \text{Line 3} \end{array}$$

2. Vertex Intersection (Cramer's Rule)

For lines L_1, L_2 ($D = a_{11}a_{22} - a_{21}a_{12} \neq 0$):

$$x = \frac{b_1a_{22} - b_2a_{12}}{D}, \quad y = \frac{a_{11}b_2 - a_{21}b_1}{D}$$

Gives corners P_{12}, P_{23}, P_{31} ($D = 0 \implies$ parallel).

3. Triangle Properties

Area (Shoelace):

$$\Delta = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

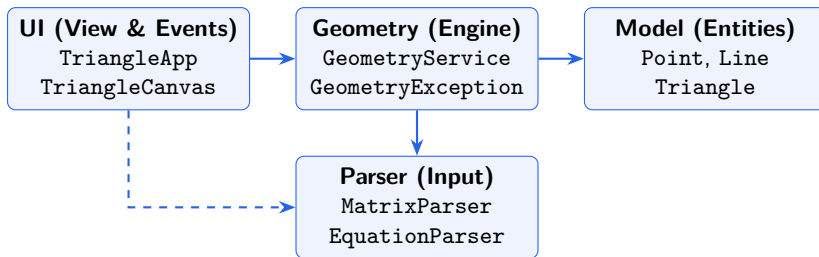
Centroid (Balance): $G = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3} \right)$

Perimeter: $p = a + b + c$ ($d = \sqrt{\Delta x^2 + \Delta y^2}$)

4. Geometric Validation

Valid $\iff D \neq 0$, distinct corners, and $\Delta \geq 10^{-6}$.
(Rejects parallel, coincident, concurrent, or flat lines)

4. The Solution: Layered Architecture



Architecture Principles

- **Decoupled Core:** Math in model/geometry has zero JavaFX UI dependencies.
- **Separation of Concerns:** UI renders; Parser sanitizes; Geometry validates.

Key Engineering Benefits

- **Headless Testing:** All math is 100% testable with JUnit without a display.
- **Clear Failure Feedback:** Custom exceptions explain invalid geometry in the UI.

5. Solution Walk-through: Validation Pipeline

Fail-Fast Validation Pipeline:

- ➊ **Pairwise check:** Reject parallel ($D = 0$) or coincident lines.
- ➋ **Cramer's intersection:** Compute vertices (P_{12}, P_{23}, P_{31}).
- ➌ **Concurrency check:** If $P_{12} \approx P_{23}$, lines cross at 1 point ($\epsilon = 10^{-6}$).
- ➍ **Collinearity / Area:** Reject flat triangle if $\Delta < 10^{-6}$.
- ➎ Return immutable Triangle.

Defensive Engineering

Eliminates division-by-zero ($D = 0$) early; tolerances protect against IEEE-754 precision limits.

```
public Triangle buildTriangle(Line l1, Line l2, Line l3)
    throws GeometryException {
    // 1. Fail-fast: reject parallel / identical pairs
    checkPair(l1, l2, 1, 2);
    checkPair(l2, l3, 2, 3);
    checkPair(l3, l1, 3, 1);

    // 2. Pairwise intersection via Cramer's rule
    Point p12 = l1.intersectionWith(l2).orElseThrow();
    Point p23 = l2.intersectionWith(l3).orElseThrow();
    Point p31 = l3.intersectionWith(l1).orElseThrow();

    // 3. Concurrency check (tolerance = 1e-6)
    if (p12.equalsWithTolerance(p23, TOL) ||
        p23.equalsWithTolerance(p31, TOL) ||
        p31.equalsWithTolerance(p12, TOL)) {
        throw new GeometryException("Lines are concurrent.");
    }

    // 4. Area threshold check (reject degenerate/flat)
    Triangle t = new Triangle(p12, p23, p31, l1, l2, l3);
    if (t.getArea() < MIN_AREA) {
        throw new GeometryException("Collapsed triangle.");
    }
    return t;
}
```

6. Rendering: World-to-Screen Mapping

Window-to-Viewport Transformation

- **World Space:** (0,0) Cartesian center, +y points **UP**
- **Screen Space:** (0,0) Top-Left corner, +y points **DOWN**

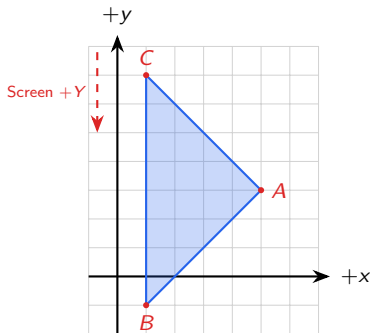
$$X_{px} = \frac{W}{2} + (x - x_c) \cdot s, \quad Y_{px} = \frac{H}{2} - (y - y_c) \cdot s$$

Uniform Scaling & Margins (Aspect 1:1):

$$s = \min \left(\frac{W - 80}{\text{span}_x \cdot 1.7}, \frac{H - 80}{\text{span}_y \cdot 1.7} \right)$$

Painter's Algorithm Pipeline

Grid → Axes → Extended Dashed Lines → Alpha Fill ($\alpha = 0.25$) → Stroke → Vertex Badges (radial offset).



Vertices: A(5, 3), B(1, -1), C(1, 7) (radial label offset from G)

7. Design Patterns Used

Facade

Class: GeometryService

Method: buildTriangle(...)

The UI calls this one method. It handles equation parsing, line intersections, and triangle validation.

Observer

Class: TriangleCanvas

Observed values: canvas width and height

When the window size changes, JavaFX notifies the canvas. The canvas then calls `redraw()`.

Call from the UI

```
service.buildTriangle(eq1, eq2,  
eq3)
```

Listener in the code

```
widthProperty().addListener(...)  
heightProperty().addListener(...)
```


8. Tests

GeometryServiceTest

- Builds a triangle from equations
- Builds a triangle from a matrix
- Rejects invalid matrix sizes
- Rejects parallel, identical, and concurrent lines
- Checks a right triangle

What these tests check

The geometry service returns the expected points and measurements for valid input, and reports an error for invalid input.

EquationParserTest

- Reads standard equations
- Reads vertical and horizontal lines
- Reads decimal and negative values
- Rejects invalid equations

MatrixParserTest

- Reads both supported matrix formats
- Rejects the wrong number of values

9. Reference Material, Lessons and Next Steps

Reference books (Books/)

- *Core Java, Vol. 1 & 2*: classes, immutability, exceptions, regular expressions
- *Introducing JavaFX 8 Programming* (Weaver): scene graph, Canvas, property listeners
- *Graphics with Java* (2007): 2D drawing and coordinate transformations
- *Digital Image Processing* (Burger & Burge): geometric coordinate mapping

Lessons learned

- Fix the input model first
- Floating point needs explicit tolerances
- Domain exceptions give clear UI messages

Next steps

- Property-based tests
- Export drawing as PNG/SVG
- Generalise to n lines

Thank you — Questions?